

LAB ASSIGNMENT 3

QUES 1 : Write a program to implement a linear regression model on the given dataset and find R2 score.

CODE :

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

da = pd.read_csv("D:/IML lab/data.csv")
print(da.head())

da["Date"] = pd.to_datetime(da["Date"])

da["Date"] = (da["Date"] - da["Date"].min()).dt.days

X = da[["Date"]]

y = da["Daily minimum temperatures"]

X_train , X_test , y_train , y_test = train_test_split(
    X,y,test_size = 0.2 , random_state = 42
)
model = LinearRegression()
model.fit(X_train,y_train)

y_pred = model.predict(X_test)

r2 = r2_score(y_test , y_pred)
print("R2 Score :", r2)
```

OUTPUT :

```
PS D:\IML lab> python -u "d:\IML lab\Lab_4\2.py"
'      Date  Daily minimum temperatures
0  1/1/1981                20.7
1  1/2/1981                17.9
2  1/3/1981                18.8
3  1/4/1981                14.6
4  1/5/1981                15.8
R2 Score : -0.004938694958721346
```

QUES 2 : Write a program to implement a linear regression model on the given dataset and find the R2 score. (HARDCODE).

CODE :

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
class LR:
    def fit(self, X, y):
        X = np.array(X).flatten()
        y = np.array(y)
        xm, ym = X.mean(), y.mean()
        self.m = np.sum((X - xm)*(y - ym)) / np.sum((X - xm)**2)
        self.b = ym - self.m * xm

    def predict(self, X):
        return self.m * np.array(X).flatten() + self.b

    def r2(self, y, y_hat):
        return 1 - np.sum((y - y_hat)**2) / np.sum((y - y.mean())**2)

df = pd.read_csv("data.csv")
df["Date"] = pd.to_datetime(df["Date"], errors="coerce")
df = df.dropna()
df["d"] = (df["Date"] - df["Date"].min()).dt.days
X = df[["d"]]
y = df["Daily minimum temperatures"]
Xtr, Xte, ytr, yte = train_test_split(X, y, test_size=0.4, random_state=42)
model = LR()
model.fit(Xtr, ytr)
pred = model.predict(Xte)
print("R2 Score:", model.r2(yte, pred))
```

OUTPUT :

```
PS D:\IML lab> python -u "d:\IML lab\Lab_4\6.PY"
● R2 Score: 7.240919363016296e-05
```

QUES 3 : Write a program to implement a Multiple linear regression model on the given dataset and find the R2 score.

CODE :

```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
df = pd.read_csv("newdata.csv")
df.columns = ["Date", "Temp", "Humidity"]
df["Date"] = pd.to_datetime(df["Date"], errors="coerce")
df = df.dropna()
df["Date"] = (df["Date"] - df["Date"].min()).dt.days
X = df[["Date", "Humidity"]]
y = df["Temp"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
r2 = r2_score(y_test, y_pred)
print("R2 Score:", r2)
```

OUTPUT :

```
PS D:\IML lab> python -u "d:\IML lab\Lab_4\4.py"
R2 Score: 1.0
PS D:\IML lab>
```

QUES 4 : Write a program to implement a Multiple linear regression model on the given dataset and find the R2 score. (HARDCODE).

CODE :

```
import pandas as pd
import numpy as np

class LR:
    def fit(self, X, y):
        X = np.c_[np.ones(len(X)), X] # add intercept
        self.coef = np.linalg.pinv(X) @ y

    def predict(self, X):
        X = np.c_[np.ones(len(X)), X]
        return X @ self.coef

    def r2(self, y, y_pred):
        return 1 - np.sum((y - y_pred)**2) / np.sum((y - y.mean())**2)

df = pd.read_csv("newdata.csv")
df.columns = df.columns.str.strip()
df["Date"] = pd.to_datetime(df["Date"], errors="coerce")
df = df.dropna()
df["t"] = (df["Date"] - df["Date"].min()).dt.days
X = df[["t", "Humidity"]].values
y = df["Daily minimum temperatures"].values
model = LR()
model.fit(X, y)
y_pred = model.predict(X)
print("R2 Score:", model.r2(y, y_pred))
```

OUTPUT :

```
PS D:\IML lab> python -u "d:\IML lab\Lab_4\5.PY"
R2 Score: 1.0
```